SHARP AND UNIFORM BOUNDS FOR THE WEAK-REGULAR ZERO-SUM INVARIANT IN RANK TWO

WENKAI YANG

Editorial note. Complete original author mathematical body and embedded bibliography retained. Authors own correctness; no independent proof certification or mathematical repair.

Editorial note. Unused bbm package omitted in derived source0; no mathbbm command occurs in original body. Raw TeX retained.

Editorial note. Unified c0 theorem is counted once, including its decomposition and Kneser arguments; intro split-case consequences not extra targets. N(G) target bundles original lower-bound construction and every necessary cyclic/index-p/quotient lemma.

Editorial note. Original acknowledgement says material is based on parts of author doctoral dissertation completed in2025; this provenance remains in source context.

Allowed standard prerequisites (editorial).

- Finite abelian sumsets, regular/weak-regular sequences and Kneser theory. All original sequence multiplicity conventions, additive-basis and N(G) definitions, quotient and subgroup classification, cited base values and supporting author proofs retained.. Complete original mathematical body and embedded bibliography; prime-minimality and odd-order hypotheses unchanged.

1. Introduction and main results

Let G be a finite abelian group, with additive notation, and let S be a sequence over G. Denote by $\sum(S)$ the set of sums of nonempty subsequences of S. The sequence S is an additive basis of G if $\sum(S) = G$. For a subgroup $H \le G$, let S_H be the subsequence consisting of all terms of S in H. We call S regular over G if $|S_H| \le |H| - 1$ for every proper subgroup $H < G$. Let $c_0(G)$ be the least integer l such that every regular sequence over G of length at least l is an additive basis. The problem of determining $c_0(G)$ was proposed by Olson; see [2, 6].

Let p be the smallest prime divisor of $|G|$, and let

$$m(G) = \begin{cases} |G|, & G \text{ is cyclic}, \\ 2p-1, & G = C_p^2, \\ np+2p-3, & G = C_p \oplus C_{np},\ n \ge 2, \\ |G|/p+p-2, & \text{otherwise}. \end{cases}$$

The inequality $c_0(G) \ge m(G)$ is known for every finite abelian group [6, Lemma 2.3]. Equality has been established for many groups of rank at least three; see [2, 6]. For a group $G = C_{n_1} \oplus C_{n_2}$ with $1 < n_1 \mid n_2$, the known cases include the following:

(1) $|G|$ is even, or G is a p-group [2, 10];
(2) $G = C_3 \oplus C_{3q}$, where $q \ge 5$ is prime [11];
(3) $p \ge 3$, $n_1 \ge 2p$ and $|G| \ge 72p^6$ [5];
(4) $p \ge 11$ and $n_1 \ge p^2$ [12].

Date: arxiv:2610.13103v1; CC BY 4.0.

Qu, Li, Wang and Zhao also proved that $c_0(G) \le 2|G|/5+1$ for every group of rank two with $p \ge 11$ [12, Lemma 7]. For $G = C_p \oplus C_{np}$ with $n \ge 2$, Qu and Han conjectured that $c_0(G) = np + 2p - 3$ [10, Concluding Remark].

We also study the invariant $\mathsf{N}(G)$ introduced by Gao et al. [3]. A sequence S over G is weak-regular if $\mathsf{v}_g(S) \le \mathrm{ord}(g)$ for every $g \in G$, where $\mathsf{v}_g(S)$ denotes the multiplicity of g in S. The invariant $\mathsf{N}(G)$ is the least integer l such that every weak-regular sequence S over G with $|S| \ge l$ has a nonempty zero-sum subsequence T satisfying $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$. Equivalently, $\mathrm{supp}(ST^{-1}) \ne \mathrm{supp}(S)$. Gao et al. determined $\mathsf{N}(C_p^r) = r(p-1)+1$ for $p \le 3$ and $\mathsf{N}(C_p^2) = 4p - 2\lceil 2\sqrt{p+1}\rceil + 1$ for $p \ge 3$ [4, Theorem 1.3]. For cyclic groups, Jiang and the author proved

$$2n - \lceil 2\sqrt{n+1}\rceil \le \mathsf{N}(C_n) \le 2n - \lceil 2\sqrt{n+1}\rceil + 1 \qquad (n \ge 3),$$

with each equality holding for infinitely many n [8, Theorem 1.1].

Our bounds for additive bases follow from the estimate

$$c_0(C_q \oplus C_{nq}) \le \max\left\{ nq + 2q^2 - q - 1,\ \frac{nq(q+1)}{p} - q + p - 2 \right\},$$

where q is prime, $n \ge 2$, and $p \ge 3$ is the smallest prime divisor of nq; see Theorem 3.5. This estimate applies whether or not $q \mid n$. The first two theorems give consequences for the two cases separately.

Theorem 1.1. *Let q be prime and let $G = C_q \oplus C_{nq^2}$, where $n \ge 1$ is not a power of q. If $p \ge 3$ is the smallest prime dividing $|G|$, then*

$$c_0(G) \le \max\{\frac{nq^2(q+1)}{p}, nq^2 + f(q) - 1\},$$

where $f(q) = 4q(q-1)+1$ for $q \ge 7$, $f(q) = 2q^2+3$ for $q \le 5$.

Theorem 1.2. *Let q be prime and let $G = C_q \oplus C_{nq}$, where $n \ge 1$ and $q \nmid n$. If $p \ge 3$ is the smallest prime dividing $|G|$, then*

$$c_0(G) \le \max\{(2q-1)(n+q+1), \frac{nq^2}{p} + 2\frac{nq}{p}\}.$$

For $\mathsf{N}(G)$, we prove the lower bound $r(n_r - 1) + 1$. The other term in the following theorem is the direct-sum bound of Gao et al. [4, Theorem 1.1(2)].

Theorem 1.3. *Let $G = C_{n_1} \oplus \cdots \oplus C_{n_r}$, where $1 < n_1 \mid \cdots \mid n_r$. Then*

$$\mathsf{N}(G) \ge \max\left\{ r(n_r - 1) + 1,\ \sum_{i=1}^{r} \big(\mathsf{N}(C_{n_i}) - 1\big) + 1 \right\}.$$

Theorem 1.4. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Then*

$$2np - 1 \le \mathsf{N}(G) \le \begin{cases} 2np - 1, & p \mid n, \\ 5np - 4, & p \ge 5,\ p \nmid n, \\ 4np - 3, & p = 3,\ 3 \nmid n. \end{cases}$$

In particular, $\mathsf{N}(C_p \oplus C_{np^2}) = 2np^2 - 1$ whenever $n \ge 1$ and $p \ge 3$ is the smallest prime dividing the group order. The two upper bounds for $p \nmid n$ also hold when $n = 1$.

Section 2 gives the notation and preliminary lemmas. Section 3 proves Theorems 1.1 and 1.2. Section 4 proves Theorems 1.3 and 1.4. The final section discusses the remaining rank-two cases and two cyclic families.

2. Notation and preliminaries

Our notation for sequences follows the standard conventions of zero-sum theory; see [1, 7]. A sequence over G is finite and unordered, and repetitions are allowed. For a sequence $S = g_1 \cdots g_l$, its length is $|S| = l$, its sum is $\sigma(S) = g_1 + \cdots + g_l$, and its support is $\mathrm{supp}(S) = \{g \in G : \mathsf{v}_g(S) > 0\}$. The relation $T \mid S$ means that T is a subsequence of S; then ST^{-1} is the subsequence obtained by deleting T from S. The empty sequence is denoted by 1 and has sum 0. Two subsequences T_1, T_2 of S are disjoint if $T_1T_2 \mid S$. A homomorphism $\varphi : G \to G_1$ induces the sequence $\varphi(S) = \varphi(g_1) \cdots \varphi(g_l)$ over G_1.

Let $\sum_0(S) = \{0\} \cup \sum(S)$. Thus $\sum_0(S)$ includes the sum of the empty subsequence, whereas $\sum(S)$ consists of sums of nonempty subsequences. A nonempty sequence is a minimal zero-sum sequence if it has sum 0 and no proper nonempty zero-sum subsequence. The Davenport constant $\mathsf{D}(G)$ is the least integer l such that every sequence over G of length at least l has a nonempty zero-sum subsequence. We use $\mathsf{D}(C_{n_1} \oplus C_{n_2}) = n_1 + n_2 - 1$ for $1 \le n_1 \mid n_2$ [7, Theorem 5.8.3]. Let $\mathbb{N}$ denote the positive integers. We use the usual floor and ceiling functions $\lfloor x \rfloor$ and $\lceil x \rceil$.

For each subset A of G, denote by $\mathrm{st}(A) = \{g \in G : g + A = A\}$ the stabilizer of A in G. Then $\mathrm{st}(A)$ is the maximal subgroup H of G such that $H + A = A$. We use the following form of Kneser's theorem; see [9, Theorem 4.4].

Lemma 2.1 (Kneser)**.** *Let $A_1, ..., A_n$ be nonempty finite subsets of an abelian group G, and $H = \mathrm{st}(A_1 + \cdots + A_n)$ be the stabilizer. Then*

$$|A_1 + \cdots + A_n| \ge |A_1 + H| + \cdots + |A_n + H| - (n-1)|H|.$$

Lemma 2.2 (Cauchy–Davenport)**.** *Let $A_1, ..., A_n$ be nonempty finite subsets of an abelian group $G = C_p$, where p is prime. Then*

$$|A_1 + \cdots + A_n| \ge \min\{p, |A_1| + \cdots + |A_n| - (n-1)\}.$$

Lemma 2.3. *For every finite abelian group G,*

$$m(G) \le c_0(G) \le |G|.$$

See [6, Lemma 2.3] for the lower bound and [2, Lemma 2.4] for the upper bound.

Qu and Han determined $c_0(G)$ for $G = C_p \oplus C_{p^n}$ as follows.

Lemma 2.4 ([10])**.** *Let $G = C_p \oplus C_{p^n}$, where p is prime and $n \ge 2$. Then*

$$c_0(G) = p^n + 2p - 3.$$

Lemma 2.5 ([2, Lemma 2.3(1)])**.** *Let G be a finite abelian group, and let p be the smallest prime dividing $|G|$. Let S be a regular sequence over G of length $|S| \ge \max\{|G|/p + p - 2,\ \mathsf{D}(G)\}$. If $\sum(S) \ne G$, then $\mathrm{st}(\sum(S)) = \{0\}$.*

We use the cyclic bound

$$\mathsf{N}(C_n) \le 2n - \lceil 2\sqrt{n+1} \rceil + 1 \qquad (n \ge 3)$$

from [8, Theorem 1.1], and the value $\mathsf{N}(C_p^2) = 4p - 2\lceil 2\sqrt{p+1} \rceil + 1$ for $p \ge 3$ from [4, Theorem 1.3(3)]. Also $\mathsf{N}(C_2) = 2$.

Lemma 2.6. *Let $G = C_p \oplus C_{np}$, where $p \ge 3$ is the smallest prime divisor of $|G|$. For $1 < m \mid n$, let $H \le pG$ have order m, and let $k = n/m$. Then*

$$2np - 1 - \big(\mathsf{N}(H) - 1 + (m-1)kp\big) \ge \begin{cases} 2kp - 1, & p \mid k,\ m \ne n, \\ 4kp - 3, & p = 3,\ p \nmid k,\ m \ne n, \\ 5kp - 4, & p \ge 5,\ p \nmid k,\ m \ne n, \\ \mathsf{N}(C_p^2), & m = n. \end{cases}$$

Proof. Since $m \ge p \ge 3$, the cyclic bound gives $\mathsf{N}(H) \le 2m-3$. Therefore

$$\begin{aligned} 2np - 1 - \big(\mathsf{N}(H) - 1 + (m-1)kp\big) &= kp(m+1) - \mathsf{N}(H) \\ &\ge kp(m+1) - 2m + 3. \end{aligned}$$

For $m < n$, subtracting $2kp-1$ from the last expression gives $(kp-2)(m-1)+2 \ge 0$. When $p=3$, subtracting $4kp-3$ gives $(m-3)(kp-2) \ge 0$. When $p \ge 5$, subtracting $5kp-4$ gives $(m-4)kp - 2m + 7 \ge 3m - 13 > 0$. These prove the first three estimates. If $m = n$, then $k = 1$, and the left side is at least

$$(p-2)n + p + 3 \ge p^2 - p + 3 > 4p - 7 \ge \mathsf{N}(C_p^2).$$

The strict inequality follows from $p^2 - 5p + 10 = (p-2)(p-3) + 4 > 0$. □

Lemma 2.7. *Let S be a weak-regular sequence over $G = C_n$, where $n \ge 2$ and $|S| \ge n$. Then either $\sum(S) = G$, or there is a zero-sum subsequence $T \mid S$ such that $\mathsf{v}_g(T) = \mathsf{v}_g(S) \ge 1$ for some $g \mid S$.*

Proof. If $\mathsf{v}_g(S) = \mathrm{ord}(g)$ for some g, take $T = g^{\mathrm{ord}(g)}$. Otherwise $0 \notin \mathrm{supp}(S)$, and we may assume $\sum(S) \ne G$. Since $|S| \ge \mathsf{D}(G)$, $\sum(S) = \sum_0(S)$. Let $H = \mathrm{st}(\sum_0(S))$ and $m = |H|$. If $m = 1$, Kneser's theorem gives $|\sum_0(S)| \ge |S| + 1 > n$. Thus $m > 1$, and $\sum_0(S) \ne G$ gives $m < n$. Since $n = dm$ for some integer $d \ge 2$, Kneser's theorem also gives

$$(|S| - |S_H| + 1)m \le \Big|\sum_0(S)\Big| \le n - m, \qquad |S_H| \ge n - d + 2 \ge 2m \ge \mathsf{N}(H).$$

The middle inequality is equivalent to $(d-2)(m-1) \ge 0$. Apply the definition of $\mathsf{N}(H)$ to the weak-regular sequence S_H; all occurrences in S of the resulting element $g \in H$ belong to S_H. □

Lemma 2.8. *Let S, S' be sequences over a finite abelian group G, and let $H \le G$ satisfy $H \subseteq \sum_0(S)$. Suppose that $T_1 \mid S'$ satisfies $\sigma(T_1) \in H$ and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S') \ge 1$ for some $g \in \mathrm{supp}(S') \setminus \mathrm{supp}(S)$. Then SS' has a zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(SS')$.*

Proof. Choose $T_2 \mid S$ with $\sigma(T_2) = -\sigma(T_1)$ and take $T = T_1T_2$. Since $g \notin \mathrm{supp}(S)$, its multiplicity in T equals that in SS'. □

3. Proof of Theorems 1.1 and 1.2

Lemma 3.1. *Let S be a regular sequence over a finite abelian group G. Then*

$$|\sum_0(S)| \ge \min\{|G|, |S|+1\}.$$

In particular, $\sum_0(S) = G$ if $|S| \ge |G| - 1$.

Proof. Suppose that $|\sum_0(S)| < \min\{|G|, |S|+1\}$. Let $H = \mathrm{st}(\sum_0(S))$ and $m = |H|$. Then $H < G$ and $m \le |S|$, so Kneser's theorem gives

$$|\sum_0(S)| \ge (|S| - |S_H| + 1)m \ge (|S| - m + 2)m \ge |S| + 1,$$

a contradiction. □

Lemma 3.2. *Let $K < H < G$, where $|K| \ge 2$ and $H/K \cong C_q$ for a prime q. Let S be a sequence over $H \setminus K$ that is regular over G. If $|S| \ge 2q-1$, then there exists $T \mid S$ such that $|T| \le q$ and $\sigma(T) \in K \setminus \{0\}$.*

Proof. Take $RR' \mid S$ with $|R| = q$ and $|R'| = q-1$. Since R is regular over H and $|H| = q|K| \geq 2q$, Lemma 3.1 gives $|\sum_0(R)| \geq q+1$. Thus there exist $R_1, R_2 \mid R$ whose sums are distinct and belong to the same coset of K. By Cauchy–Davenport, $\sum_0(R') + K = H$. Choose $R_3 \mid R'$ such that $\sigma(R_1R_3), \sigma(R_2R_3) \in K$. Choose $U \in \{R_1R_3, R_2R_3\}$ with $\sigma(U) \neq 0$. Factor U into subsequences whose images in H/K are minimal zero-sum sequences. Each factor has length at most q, and at least one has nonzero sum, as required. □

Lemma 3.3. *Let $K \leq G$ with $|K| \geq 2$ and $G/K \cong C_q^2$ for a prime q. If S is regular over G and*

$$|S| \geq q|K| + 2q^2 - q - 1,$$

then $S = S_K T_1 \cdots T_k R$, where $k = |K| - |S_K|$, $|T_i| \leq q$, $\sigma(T_i) \in K \setminus \{0\}$ and $\sum_0(R) + K = G$.

Proof. Let $\varphi : G \to G/K$ be the natural homomorphism and let $k = |K| - |S_K|$. Starting with SS_K^{-1}, choose at each step a subgroup of order q containing the largest number of remaining images. Before each of the first k steps, the remaining length is at least $|S| - |S_K| - q(k-1) \geq 2q^2 - 1$, so the chosen subgroup contains at least $2q-1$ terms. Apply Lemma 3.2 in its inverse image to obtain $T_1, \ldots, T_k$. The remaining sequence $R = S(S_K T_1 \cdots T_k)^{-1}$ has length $|R| \geq 2q^2 - q - 1$. Suppose that $\sum_0(\varphi(R)) \neq G/K$. Since $|R| > (q+1)(q-2)$, some subgroup $\bar{H}$ of order q contains at least $q-1$ terms of $\varphi(R)$. For $H = \varphi^{-1}(\bar{H})$, Cauchy–Davenport gives $\sum_0(\varphi(R_H)) = \bar{H}$. It also gives $|RR_H^{-1}| \leq q-2$, since otherwise $\sum_0(\varphi(R)) = G/K$. Thus $|R_H| \geq 2q^2 - 2q + 1$. Working backwards, suppose that $T_{j+1}, \ldots, T_k$ lie in H. Before T_j was chosen, at most $q-2+q = 2q-2$ terms lay outside H, whereas at least $2q^2 - 2q + 1$ lay in H. Distinct subgroups of order q in G/K meet only at zero, and the remaining terms have nonzero images. Thus the subgroup containing the largest number of remaining images is $\bar{H}$, forcing T_j to lie in H. Hence all T_i lie in H, and $|S_H| \geq |S| - q + 2 > |H| = q|K|$, a contradiction. □

Lemma 3.4. *Let p be the smallest prime dividing $|G|$, let $K < G$ with $|K| \geq 2$, and let $l \geq 2$ be an integer. Suppose that S is regular over G and admits a decomposition $S = S_K T_1 \cdots T_k R$, where $k = |K| - |S_K|$, $|T_i| \leq l$, $\sigma(T_i) = t_i \in K \setminus \{0\}$ and $\sum_0(R) + K = G$. If*

$$|S| \geq \frac{|G|}{p} + l\frac{|K|}{p} - l + p - 2,$$

then $\sum(S) = G$.

Proof. Let $W = S_K t_1 \cdots t_k$. Since $|W| = |K| \geq \mathsf{D}(K)$, we have $0 \in \sum(S)$. If W is regular over K, then $\sum_0(W) = K$ and the conclusion follows. Otherwise choose a minimal subgroup $H_1 < K$ with $|W_{H_1}| \geq |H_1| = m$. Then $H_1 \neq \{0\}$ and W_{H_1} is regular over H_1. Let $r = m - |S_{H_1}| - 1$. After reordering, $S_{H_1} t_1 \cdots t_r$ is regular over H_1 of length $m-1$. Its subsequence sums together with 0 therefore cover H_1 by Lemma 3.1. Let $W_0 = S t_1 \cdots t_r (T_1 \cdots T_r)^{-1}$ and $H_2 = \mathrm{st}(\sum_0(W_0))$. Since the T_i are disjoint, replacing each selected occurrence of t_i by T_i shows that $\sum_0(W_0) \subseteq \sum_0(S)$. Together with $0 \in \sum(S)$, this gives $\sum_0(W_0) \subseteq \sum(S)$. The subsequence $S_{H_1} t_1 \cdots t_r$ of W_0 has $\sum_0(S_{H_1} t_1 \cdots t_r) = H_1$, so $\sum_0(W_0)$ is a union of cosets of H_1. Hence $H_1 \leq H_2$. We may assume $H_2 < G$; hence $p \leq |H_2| \leq |G|/p$. Since $r \leq |K|/p - 1$, $|W_0| \geq |S| - (l-1)r$ and $|(W_0)_{H_2}| \leq |S_{H_2}| + r \leq |H_2| - 1 + r$, Kneser's theorem gives

$$|\sum_0(W_0)| \geq (|S| - lr + 2 - |H_2|)|H_2| \geq (|G|/p + p - |H_2|)|H_2| \geq |G|.$$

The last inequality follows from $(|H_2| - p)(|G|/p - |H_2|) \geq 0$, and thus $\sum(S) = G$. □

Theorem 3.5. *Let $G = C_q \oplus C_{nq}$, where q is prime and $n \geq 2$, and let $p \geq 3$ be the smallest prime dividing $|G|$. Then*

$$c_0(G) \leq \max\left\{nq + 2q^2 - q - 1,\ \frac{nq(q+1)}{p} - q + p - 2\right\}.$$

Proof. Let S be regular over G with length at least the stated bound. For $K = qG$, we have $|K| = n$ and $G/K \cong C_q^2$. The first term of the bound allows us to apply Lemma 3.3. The second term is the bound in Lemma 3.4 with $l = q$. Hence $\sum(S) = G$. □

Proof of Theorem 1.1. Apply Theorem 3.5 with nq in place of n. It gives

$$c_0(G) \leq \max\left\{nq^2 + 2q^2 - q - 1,\ \frac{nq^2(q+1)}{p} - q + p - 2\right\}.$$

Since $p \leq q$, the second term is at most $nq^2(q+1)/p$. For the first, $f(q) - 2q^2 + q$ equals $2q^2 - 3q + 1$ when $q \geq 7$ and $q + 3$ when $q = 3, 5$, and is positive in both cases. This proves the stated bound. □

Proof of Theorem 1.2. If $n = 1$, then $G = C_q^2$ and $c_0(G) = 2q - 1$, which is at most $(2q-1)(n+q+1)$. Suppose that $n \geq 2$. Since $p \geq 3$ is the smallest prime dividing $|G|$, Theorem 3.5 applies and gives

$$c_0(G) \leq \max\left\{nq + 2q^2 - q - 1,\ \frac{nq(q+1)}{p} - q + p - 2\right\}.$$

The first is at most $(2q-1)(n+q+1)$, since subtracting $nq+2q^2-q-1$ from that expression gives $(q-1)n+2q > 0$. For the second, $p \leq q$ implies $nq(q+1)/p-q+p-2 \leq nq(q+1)/p$. The latter is at most $nq^2/p + 2nq/p$, because their difference is $nq/p > 0$. Taking the maximum of these two comparisons proves the result. □

4. Proof of Theorems 1.3 and 1.4

Proof of Theorem 1.3. Choose a basis $(x_1, \ldots, x_r)$ with $\operatorname{ord}(x_i) = n_i$ and let $g_0 = x_r$, $g_i = x_r + x_i$ for $1 \leq i < r$. All g_i have order n_r, so $S = \prod_{i=0}^{r-1} g_i^{n_r-1}$ is weak-regular. For a zero-sum subsequence $T = \prod_{i=0}^{r-1} g_i^{a_i}$, the coordinates give $a_i \equiv 0 \pmod{n_i}$ for $i \geq 1$ and $\sum_i a_i \equiv 0 \pmod{n_r}$. Thus also $a_0 \equiv 0 \pmod{n_1}$, and none of the a_i equals $n_r - 1$. This proves $\mathsf{N}(G) \geq r(n_r - 1) + 1$. For completeness, consider the direct-sum bound from [4, Theorem 1.1(2)]. For each i, choose a weak-regular sequence over $\langle x_i \rangle$ of length $\mathsf{N}(C_{n_i}) - 1$ having no nonempty zero-sum subsequence that contains all occurrences of any of its elements. None contains 0, and the direct factors intersect only at 0, so their product in G is weak-regular. In a direct sum, a sequence has sum zero if and only if its sum in every coordinate is zero. A zero-sum subsequence containing all occurrences of some element would therefore have the same property in the corresponding factor, contrary to the choice of the sequences. The product has length $\sum_{i=1}^{r}(\mathsf{N}(C_{n_i}) - 1)$, giving the second bound. □

If $rn_r \geq 2\sum_{i=1}^{r} n_i$, the first lower bound dominates the second, since $\mathsf{N}(C_{n_i}) - 1 \leq 2n_i - 2$.

Lemma 4.1. *Let $G = C_p \oplus C_{np}$, where $n \geq 2$ and $p \geq 3$ is the smallest prime dividing $|G|$. Then*

$$c_0(G) \leq \begin{cases} \lceil (4np-2)/3 \rceil, & p \mid n, \\ \lceil (25np-20)/6 \rceil, & p \geq 5,\ p \nmid n, \\ \lceil (16np-12)/5 \rceil, & p = 3,\ 3 \nmid n. \end{cases}$$

Proof. Suppose first that $p \mid n$. If n is a power of p, then $c_0(G) = np + 2p - 3 \le \lceil (4np-2)/3 \rceil$, since $np \ge p^2 \ge 6p-9$. Otherwise $n \ge p(p+2)$, whence $np \ge p^2(p+2) \ge 6p^2 - 3p - 1$. Thus

$$np + 2p^2 - p - 1 \le \frac{4np-2}{3}, \qquad n(p+1) - 2 \le \frac{4np-2}{3}.$$

Theorem 3.5 proves the first assertion. If $p \nmid n$, then $n \ge p+2$, and the same theorem gives $c_0(G) \le 3np$. For $p \ge 5$, this is at most $\lceil (25np-20)/6 \rceil$; for $p = 3$, it is at most $\lceil (16np-12)/5 \rceil$, since $np \ge 15$. □

Proposition 4.2. *Let $G = C_p \oplus C_{np}$, where p is prime and $n \ge 1$, and let l_0 be a positive integer. Let $r = \lceil l_0/(np-1) \rceil$. If*

$$l_0 - \lfloor l_0/r \rfloor \ge c_0(G),$$

then every regular sequence S over G with $|S| \ge l_0$ has a nonempty zero-sum subsequence T satisfying $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$.

Proof. Since $\langle g \rangle < G$ for every $g \in G$, regularity gives $\mathsf{v}_g(S) \le \mathrm{ord}(g) - 1 \le np - 1$. Hence $|\operatorname{supp}(S)| \ge r$. Choose $g_0 \mid S$ with minimum multiplicity $a = \mathsf{v}_{g_0}(S)$. The sequence $S' = Sg_0^{-a}$ contains none of the occurrences of g_0. Then S' is regular, $a \le \lfloor |S|/r \rfloor$, and

$$|S'| \ge |S| - \lfloor |S|/r \rfloor \ge l_0 - \lfloor l_0/r \rfloor \ge c_0(G).$$

Choose $T' \mid S'$ with $\sigma(T') = -ag_0$. Then $T = T'g_0^a$ has the required property. □

For $n \ge 2$ and $p \ge 3$ the smallest prime dividing $|G|$, let

$$\mathrm{ub}(G) = \begin{cases} 2np - 1, & p \mid n, \\ 5np - 4, & p \ge 5,\ p \nmid n, \\ 4np - 3, & p = 3,\ 3 \nmid n. \end{cases}$$

Corollary 4.3. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Every regular sequence S over G with $|S| \ge \mathrm{ub}(G)$ has a nonempty zero-sum subsequence T such that $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$.*

Proof. Apply Proposition 4.2 with $l_0 = \mathrm{ub}(G)$. The corresponding values of r are 3, 6, and 5, according as $p \mid n$, $p \ge 5$ and $p \nmid n$, or $p = 3$ and $3 \nmid n$. In each case, the required inequality is precisely Lemma 4.1. □

Lemma 4.4. *Let S be a non-regular sequence over a finite abelian group G. Then there exists a proper subgroup $H < G$ such that S_H is regular over H, $|S_H| \ge |H|$, and $\sum(S_H) = H$.*

Proof. Choose $H < G$ of minimum order subject to $|S_H| \ge |H|$. For every proper subgroup $H' < H$, minimality gives $|S_{H'}| \le |H'| - 1$. Thus S_H is regular over H. Since $c_0(H) \le |H|$, we obtain $\sum(S_H) = H$. □

Lemma 4.5. *Let $S = S_H S_1$ with $\sum_0(S_H) = H$ and $\operatorname{supp}(S_1) \subseteq G \setminus H$, and let $\Phi : G \to G/H$ be the natural homomorphism. Suppose that some $\bar{g} \in \operatorname{supp}(\Phi(S_1))$ satisfies $\mathsf{v}_{\bar{g}}(\Phi(S_1)) \ge \mathrm{ord}(\bar{g})$ and that $\Phi^{-1}(\bar{g}) \cap \operatorname{supp}(S_1)$ contains at least two elements. Then S has a nonempty zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S_1$.*

Proof. Let $V \mid S_1$ consist of all terms whose image is $\bar{g}$, and let $n = \mathrm{ord}(\bar{g})$. If $\mathsf{v}_h(V) \ge n$ for some $h \mid V$, let $|V| = kn + t$, where $0 \le t < n$, and take $T_1 = Vh^{-t}$. Then $\sigma(T_1) \in H$, and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$ for every $g \mid V$ with $g \ne h$. Such an element exists by hypothesis. Otherwise, choose $g \mid V$. Since $\mathsf{v}_g(V) < n \le |V|$, there is an n-term subsequence $T_1 \mid V$ containing all occurrences of g. Again $\sigma(T_1) \in H$ and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$. Lemma 2.8 completes the proof in both cases. □

Lemma 4.6. *Let $G = C_p \oplus C_{np}$, where $p \geq 3$ is the smallest prime dividing $|G|$, $n \geq 5$, and $H < G$ is cyclic of order np. Let $S = S_H g_0^b$ satisfy $g_0 \notin H$, $\mathrm{ord}(g_0) = np$, $b \leq np-1$, $|S| \geq 2np-1$, and $|S_H| \geq np$. Suppose that every term of S_H generates H. Choose $h \mid S_H$ with minimum multiplicity a. Let $R = S_H h^{-a}$. If $|R| \leq np-1$ and $a < np/2$, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_h(T) = \mathsf{v}_h(S)$.*

Proof. Let $b = ps_g + t_g$, and $R = \prod_{i=1}^{\ell-1} g_i^{ps_i+t_i}$, where $0 \leq t_i, t_g < p$. First suppose that $a \geq 2p$. As $\ell \leq 1 + (np-1)/a$, convexity of $x + (p-1)(np-1)/x$ on $[2p, np/2]$ gives

$$p\Big(s_g + \sum_i s_i\Big) \geq 2np - 1 - a - \ell(p-1) \geq 3np/2 - 3p \geq np - p/2.$$

Thus $s_g + \sum_i s_i \geq n$. Since $s_g \leq n-1$, choose i with $s_i > 0$. Consider $U = (pg_0)^{s_g} \prod_j (pg_j)^{s_j}$ and delete one occurrence of pg_i. The remaining sequence has at least $n-1$ terms, all generating $pH = pG$, so its $\sum_0$ equals pH by Lemma 3.1. The deleted occurrence of pg_i leaves g_i^p unused in the corresponding subsequences of $Rg_0^{ps_g}$. Since $\sum_0(g_i^p) + pH = H$, we obtain $H \subseteq \sum_0(Rg_0^{ps_g})$.

Now suppose that $a \leq 2p-1$. Then $s_g \geq n-2$ and $|R| \geq (n-2)p + 1 \geq 3p+1$. Take disjoint $U, V \mid R$ with $|U| = |V| = p$. Kneser's theorem gives $|\sum_0(U)| \geq p+1$. Hence two different sums from U have the same image in H/pH. Using $\sum_0(V) + pH = H$, extend both by the same subsequence of V to sums in pH. At least one is nonzero. Thus there is $W \mid UV$ with $\sigma(W) \in pH \setminus \{0\}$. Since pg_0 generates pH and $\sigma(W) \neq 0$, every proper subgroup of pH contains at most one term of $(pg_0)^{s_g}\sigma(W)$, and the trivial subgroup contains none. This sequence is therefore regular over pH and has length at least $n-1$, so its $\sum_0$ equals pH. Also $|RW^{-1}| \geq p+1$, and its image has all terms nonzero in H/pH. Again $H \subseteq \sum_0(Rg_0^{ps_g})$. In either case choose $T_1 \mid Rg_0^{ps_g}$ with $\sigma(T_1) = -ah$ and take $T = T_1 h^a$. □

Lemma 4.7. *Let $G = C_p \oplus C_{np}$, where $n \geq 2$ and $p \geq 3$ is the smallest prime dividing $|G|$, and let $H < G$ be cyclic of order np. Let $S = S_H g_0^b$ satisfy $g_0 \notin H$, $\mathrm{ord}(g_0) = np$, $b \leq np-1$, $|S| \geq 2np-1$, and suppose that S_H is regular over H. If $\mathrm{ord}(h') \leq n$ for some $h' \mid S_H$, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_{h'}(T) = \mathsf{v}_{h'}(S)$.*

Proof. Let $a = \mathsf{v}_{h'}(S)$, $R' = S_H(h')^{-a}$, $s = \lfloor b/p \rfloor$, and $U = R'(pg_0)^s$. Regularity gives $a \leq n-1$, while $s \leq n-1$ and $b - s \leq n(p-1)$. Hence $|U| = |S| - a - b + s \geq np$. Suppose that $\sum_0(U) \neq H$, and let $H_1 = \mathrm{st}(\sum_0(U))$ and $d = |H_1|$. Since $0 \nmid U$, Kneser's theorem implies $H_1 \neq \{0\}$. It also gives

$$|U| - |U_{H_1}| \leq np/d - 2, \qquad |U_{H_1}| \leq d - 1 + s \leq d + n - 2.$$

As $p \leq d \leq n$, $|U| \leq np/d + d + n - 4 \leq 3n - 4 < np$, a contradiction. Thus $\sum_0(U) = H \subseteq \sum_0(R'g_0^{ps})$. Choose $T_1 \mid R'g_0^{ps}$ with $\sigma(T_1) = -ah'$; then $T = T_1(h')^a$ has the required property. □

Lemma 4.8. *Let $H = C_3 \oplus C_{3m}$, where $m \geq 1$ is odd. Every regular sequence R over H of length at least $|H| = 9m$ has a nonempty zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(R)$ for some $g \mid R$.*

Proof. Every multiplicity is at most $3m-1$, so $|\mathrm{supp}(R)| \geq 4$. Choose $g \mid R$ of minimum multiplicity a and let $R' = Rg^{-a}$. Then R' is regular over H and $|R'| \geq \lceil 3|R|/4 \rceil \geq \lceil 27m/4 \rceil$. For $m \geq 5$, Theorem 3.5 gives

$$c_0(H) \leq \max\{3m + 14, 4m - 2\} \leq 6m \leq \lceil 27m/4 \rceil.$$

For $m = 1, 3$, use $c_0(C_3^2) = 5$ and $c_0(C_3 \oplus C_9) = 12$. Thus $\sum(R') = H$. Choose $T' \mid R'$ with $\sigma(T') = -ag$. Then $T = T'g^a$ is zero-sum and $\mathsf{v}_g(T) = \mathsf{v}_g(R)$. □

Lemma 4.9. *Let $H = C_m$, let R be regular over H with $|R| \geq m/2$, and let $H_0 \leq H$ be nontrivial. Then $\sum_0(R) + H_0 = H$.*

Proof. Suppose otherwise and let H_1 be the inverse image in H of the stabilizer of the image of $\sum_0(R)$ in H/H_0. For $k = |H_1|$, we have $2 \le k \le m/2$. Kneser's theorem and regularity give

$$|R| - |R_{H_1}| \le m/k - 2, \qquad |R| \le k + m/k - 3 \le m/2 - 1,$$

contrary to $|R| \ge m/2$. The last inequality follows by convexity of $x + m/x$ on $[2, m/2]$. □

Lemma 4.10. *Let $S = S_H S_1$ be a sequence over an abelian group G, where $H = C_m$, $m \ge 2$, S_H is regular over H, $|S_H| \ge m$, and every multiplicity in S_H is smaller than the corresponding element order. Suppose that pairwise disjoint nonempty subsequences of S_1 have sums forming a sequence U over H. If $\sum_0(U)$ contains a nontrivial subgroup $H_0 \le H$, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_h(T) = \mathsf{v}_h(S)$ for some $h \mid S_H$. In particular, this holds if U has no zero term, is supported in a nontrivial subgroup $K \le H$, and $|U| \ge |K| - 1$.*

Proof. The strict multiplicity bound and $|S_H| \ge m$ ensure that S_H has at least two distinct elements. Choose $h \mid S_H$ of minimum multiplicity a. Then $R = S_H h^{-a}$ has length $|R| \ge m/2$. Lemma 4.9 gives $\sum_0(R) + H_0 = H$. Choose $V \mid R$ with $\sigma(V) + ah \in H_0$. Since $H_0 \subseteq \sum_0(U)$, a subsequence of U has sum $-\sigma(V) - ah$. The corresponding pairwise disjoint subsequences of S_1 form a subsequence T_1 with the same sum. Thus $T = Vh^a T_1$ is zero-sum. Since $h \nmid R$ and $h \nmid S_1$, we also have $\mathsf{v}_h(T) = a = \mathsf{v}_h(S)$. For the last assertion, a nontrivial stabilizer of $\sum_0(U)$ is itself contained in $\sum_0(U)$. If the stabilizer is trivial, Kneser's theorem gives $|\sum_0(U)| \ge |U|+1 \ge |K|$, so $\sum_0(U) = K$. □

Lemma 4.11. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Let $S = S_H S_1$ satisfy $|S| \ge 2np-1$ and $\mathsf{v}_g(S) < \mathrm{ord}(g)$ for every $g \mid S$. Suppose that $H < G$ is cyclic of order m, that S_H is regular over H, and that $m \le |S_H| \le \mathsf{N}(H) - 1$. If G/H is cyclic of order greater than p, then S has a nonempty zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \mid S$.*

Proof. By Lemma 3.1, $\sum_0(S_H) = H$. Let $k = np/m > 1$, so that $|G/H| = pk$, and let $\Phi : G \to G/H$ be the natural homomorphism. Since $G/pG \cong C_p^2$ and $(G/H)/p(G/H) \cong C_p$, $H/(H \cap pG)$ has order p. Thus $p \mid m$ and, as H is cyclic, $H \cap pG = pH$. We have $k \ge p$ and $|S_1| \ge 2np - \mathsf{N}(C_m)$.

By Lemma 4.5, we may assume that $\Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$ contains exactly one element whenever $\mathsf{v}_{\bar{g}}(\Phi(S_1)) \ge \mathrm{ord}(\bar{g})$. For each $g \mid S_1$, let $r_g = \mathrm{ord}(\Phi(g))$ and choose $\lfloor \mathsf{v}_g(S_1)/r_g \rfloor$ disjoint copies of g^{r_g}. Let T_0 be their product, U the sequence of their sums, and $R = S_1 T_0^{-1}$. Every term of U is nonzero, since a selected subsequence g^{r_g} satisfies $r_g \le \mathsf{v}_g(S) < \mathrm{ord}(g)$. The choice of T_0 also ensures that $\Phi(R)$ is weak-regular.

Suppose that $R_1 \mid R$ satisfies $\sigma(\Phi(R_1)) = 0$ and $\mathsf{v}_{\bar{g}}(\Phi(R_1)) = \mathsf{v}_{\bar{g}}(\Phi(R))$ for some $\bar{g} \mid \Phi(R)$. Then R_1 contains every term of R whose image is $\bar{g}$. Consequently $T_1 = R_1 T_0$ has sum in H and $\mathsf{v}_h(T_1) = \mathsf{v}_h(S_1)$ for every $h \in \Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$. Lemma 2.8 gives the required subsequence of S. In particular, it suffices to prove $|R| \ge \mathsf{N}(G/H)$.

Suppose first that $m > p$, so $m \ge p^2$. By Lemma 4.10, we may assume that $\sum_0(U)$ contains no nontrivial subgroup of H. Let $U_1 = U_{pH}$ and $U_2 = U U_1^{-1}$, with $a = |U_1|$ and $b = |U_2|$. The same lemma, applied to U_1 with $K = pH$, gives $a \le m/p - 2$. Each term of U_1 is the sum of a selected subsequence of length at most pk. If a selected subsequence corresponding to a term of U_2 had length divisible by p, its sum would lie in $H \cap pG = pH$. Its length therefore divides k and is not divisible by p, so is at most k. Since its length is greater than one, $U_2 \ne 1$ implies $k \ge p + 2$.

If $b \le p - 2$, then

$$|T_0| \le k(pa + b) \le np - (p + 2)k, \qquad |R| \ge m(k - 2) + (p + 2)k \ge 2pk.$$

Here $\mathsf{N}(C_m) \le 2m$, and $m(k-2) \ge k(p-2)$ follows from $m \ge p$ and $k \ge p$. Thus $|R| \ge 2pk \ge \mathsf{N}(C_{pk})$.

If $b \ge p-1$, choose $V \mid U_2$ of length $p-1$. Its image in $H/pH = C_p$ has all terms nonzero, so $\sum_0(V)+pH = H$. The stabilizer of $\sum_0(U)$ is trivial, since otherwise it would be a nontrivial subgroup contained in $\sum_0(U)$. For every $W \mid U$, the identity $\sum_0(U) = \sum_0(W)+\sum_0(UW^{-1})$ gives $\mathrm{st}(\sum_0(W)) \le \mathrm{st}(\sum_0(U)) = \{0\}$. Kneser's theorem therefore yields $|\sum_0(U_1)| \ge a+1$. Since $\sum_0(V)$ meets every coset of pH, $|\sum_0(U_1V)| \ge p(a+1)$. Each remaining term of U_2 increases the number of subsequence sums by at least one. Otherwise that nonzero term would belong to the stabilizer of the preceding sumset, which is trivial. It follows that

$$pa+b+1 \le |\sum_0(U)| \le m-1, \qquad |T_0| \le k(pa+b) \le np-2k.$$

Hence $|R| \ge m(k-2)+2k \ge 2pk$, since $m \ge p^2$, $k \ge p+2$, and $p^2(k-2)-2k(p-1) \ge (p-2)(p^2-2) > 0$. Again $|R| \ge \mathsf{N}(C_{pk})$.

It remains to consider $m = p$. Since $pH = \{0\}$ and every term of U is nonzero, every selected subsequence has length at most k. If $|U| \ge p-2$, then $|\sum_0(U)| \ge p-1$ by Cauchy–Davenport. Choose distinct $h, x \in \mathrm{supp}(S_H)$; such a choice exists because $|S_H| \ge p$ and every multiplicity is less than p. Another application of Cauchy–Davenport gives $\sum_0(U) + \{0, x\} = H$. Since $\sum_0(U) \subseteq \sum_0(T_0)$, there is $T_1 \mid T_0x$ with $\sigma(T_1) = -\mathsf{v}_h(S)h$. Then $T = T_1h^{\mathsf{v}_h(S)}$ is zero-sum and $\mathsf{v}_h(T) = \mathsf{v}_h(S)$. We may therefore assume $|U| \le p-3$.

If $T_0 = 1$, then $R = S_1$ and

$$|R| \ge 2pk - \mathsf{N}(C_p) \ge 2pk - 2p \ge \mathsf{N}(C_{pk}).$$

The last inequality uses $pk \ge p^2$ and the cyclic bound $\mathsf{N}(C_{pk}) \le 2pk - \lceil 2\sqrt{pk+1} \rceil + 1$. Otherwise choose $g \mid S_1$ with $\mathsf{v}_g(S_1) \ge r_g$. Let $t = \mathsf{v}_g(R) < r_g \le k$ and $R' = Rg^{-t}$. By the assumption on the cosets, $\Phi(R')$ contains no occurrence of $\Phi(g)$. It is weak-regular and

$$|R'| \ge 2pk - \mathsf{N}(C_p) - (p-3)k - (k-1) = (p+2)k - \mathsf{N}(C_p) + 1 \ge pk.$$

Apply Lemma 2.7 to $\Phi(R')$. If $\sum(\Phi(R')) = G/H$, choose $R_1 \mid R'$ such that $\Phi(\sigma(R_1)) = -\mathsf{v}_g(S_1)\Phi(g)$. Then $T_1 = R_1g^{\mathsf{v}_g(S_1)}$ has sum in H and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$, so Lemma 2.8 applies. Otherwise there is $R_1 \mid R'$ with $\sigma(\Phi(R_1)) = 0$ and $\mathsf{v}_{\bar{h}}(\Phi(R_1)) = \mathsf{v}_{\bar{h}}(\Phi(R'))$ for some $\bar{h} \mid \Phi(R')$. Since $\bar{h} \ne \Phi(g)$, its multiplicity in $\Phi(R')$ equals that in $\Phi(R)$. The argument for R_1T_0 above completes the proof. □

Lemma 4.12. *Let $G = C_p \oplus C_{np}$, where $n \ge 2$ and $p \ge 3$ is the smallest prime dividing $|G|$. Let S satisfy $|S| \ge 2np-1$ and $\mathsf{v}_g(S) < \mathrm{ord}(g)$ for every $g \in \mathrm{supp}(S)$. Suppose that $H < G$ is cyclic of order np, that S_H is regular over H, and that $np \le |S_H| \le \mathsf{N}(H)-1$. Then S has a nonempty zero-sum subsequence T such that $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for some $g \in \mathrm{supp}(S)$.*

Proof. Let $S_1 = SS_H^{-1}$, and let $\Phi : G \to G/H \cong C_p$ be the quotient map. Lemma 3.1 gives $\sum_0(S_H) = H$. By Lemma 2.8, it suffices to find $T_1 \mid S_1$ with $\sigma(T_1) \in H$ and $\mathsf{v}_g(T_1) = \mathsf{v}_g(S_1)$ for some $g \mid S_1$. Since $n \ge p$ and $np \ge p^2$, the cyclic upper bound gives

$$|S_1| \ge 2np - \mathsf{N}(C_{np}) \ge \lceil 2\sqrt{np+1} \rceil - 1 \ge 2p \ge \mathsf{N}(C_p).$$

If $\Phi(S_1)$ is weak-regular, its length gives a subsequence $T_1 \mid S_1$ such that $\sigma(\Phi(T_1)) = 0$ and $\mathsf{v}_{\bar{g}}(\Phi(T_1)) = \mathsf{v}_{\bar{g}}(\Phi(S_1))$ for some $\bar{g} \mid \Phi(S_1)$. Thus $\sigma(T_1) \in H$, and T_1 contains all occurrences of every element of S_1 whose image is $\bar{g}$. Lemma 2.8 applies.

Otherwise some $\bar{g} \in G/H$ has multiplicity greater than p in $\Phi(S_1)$. If $\Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$ contains at least two elements, apply Lemma 4.5. Suppose that this intersection is $\{g\}$. If S_1 has another element $g_0 \ne g$, then $\sum_0(g^pS_H) = G$, because $g+H$ generates G/H

and $\sum_0(S_H) = H$. Choose $T' \mid g^p S_H$ with $\sigma(T') = -\mathsf{v}_{g_0}(S_1)g_0$. Since $g_0 \nmid g^p S_H$, the sequence $T = T' g_0^{\mathsf{v}_{g_0}(S_1)}$ is a zero-sum subsequence of S with $\mathsf{v}_{g_0}(T) = \mathsf{v}_{g_0}(S)$.

It remains to consider $S_1 = g_0^b$, where $b \ge p$ and $b < \mathrm{ord}(g_0) \le np$. Choose $h \mid S_H$ of minimum multiplicity a, and let $R = S_H h^{-a}$. The strict multiplicity bounds and $|S_H| \ge np$ ensure that S_H has at least two distinct elements, so $|R| \ge |S_H|/2$. If $|R| \ge np$, Lemma 3.1 gives $\sum_0(R) = H$. Choose $T_1 \mid R$ with $\sigma(T_1) = -ah$; then $T = T_1 h^a$ has $\mathsf{v}_h(T) = a = \mathsf{v}_h(S)$. Hence assume $|R| \le np - 1$.

Case 1: $|\mathrm{supp}(R)| \ge 2$. Then $2a \le |R| \le np - 1$ and $b = |S| - |R| - a \ge np - a > np/2$, forcing $\mathrm{ord}(g_0) = np$. If some $h' \mid S_H$ does not generate H, then $\mathrm{ord}(h') \le np/p = n$, so Lemma 4.7 applies. We may therefore assume that every term of S_H generates H. For $n \ge 5$, apply Lemma 4.6. The only remaining possibility is $p = n = 3$. Here Rg_0^b is regular over G: a proper subgroup containing any term of R must equal H, where $|R| \le 8$; any other proper subgroup contains no terms or only b occurrences of g_0, and in the latter case has order at least $\mathrm{ord}(g_0) > b$. Since $a \le 4$, we have $|Rg_0^b| \ge 13 \ge c_0(C_3 \oplus C_9) = 12$. Hence there is $T_1 \mid Rg_0^b$ with $\sigma(T_1) = -ah$, and $T = T_1 h^a$ has the required property.

Case 2: $|\mathrm{supp}(R)| = 1$. Then $R = f^c$ for some $f \in H$, with $a \le c \le np - 1$. Since $2c \ge |S_H| \ge np$ and $c < \mathrm{ord}(f)$, we have $\mathrm{ord}(f) = np$. Let $c = ps_f + t_f$ and $b = ps_g + t_g$, where $0 \le t_f, t_g < p$. If $a \ge (n-1)p$, then $s_f \ge n - 1$ and $s_g \ge 1$. Otherwise $a \le (n-1)p - 1$, so $c + b \ge (n+1)p$ and

$$s_f + s_g \ge \left\lceil \frac{(n+1)p - 2(p-1)}{p} \right\rceil = n.$$

Thus $s_f + s_g \ge n$ in both cases, and $s_f \ge \lfloor n/2 \rfloor \ge 1$. Consider the sequence

$$V = (pf)^{s_f - 1}(pg_0)^{s_g} \quad \text{over } pH, \qquad |V| \ge n - 1.$$

Indeed $pH = pG \cong C_n$ and pf generates this group. Since $g_0 + H$ has order p, we have $p \mid \mathrm{ord}(g_0)$, and the integer $d = \mathrm{ord}(pg_0) = \mathrm{ord}(g_0)/p$ satisfies $d > 1$ and $s_g \le d - 1$. No proper subgroup of pH contains pf; one containing pg_0 has order at least d and contains at most $s_g \le d - 1$ terms of V. Consequently V is regular over pH, and Lemma 3.1 yields $\sum_0(V) = pH$. The terms of V are sums of pairwise disjoint subsequences of $f^{ps_f} g_0^{ps_g}$, leaving p occurrences of f unused. Since $\sum_0(f^p) + pH = H$, we obtain

$$H = \sum_0(V) + \sum_0(f^p) \subseteq \sum_0(f^{ps_f} g_0^{ps_g}) \subseteq \sum_0(f^c g_0^b).$$

Choose $T_1 \mid f^c g_0^b$ with $\sigma(T_1) = -ah$. Then $T = T_1 h^a$ is nonempty and zero-sum, with $\mathsf{v}_h(T) = a = \mathsf{v}_h(S)$. □

Proof of Theorem 1.4. The lower bound for $n \ge 2$ follows from Theorem 1.3. For $n = 1$, both stated upper bounds follow from the known value of $\mathsf{N}(C_p^2)$. Fix p and use strong induction on the integers n whose prime divisors are at least p, with the known value of $\mathsf{N}(C_p^2)$ as the base $n = 1$. Let S be weak-regular over G, with $|S| \ge \mathrm{ub}(G)$. If $\mathsf{v}_g(S) = \mathrm{ord}(g)$ for some $g \mid S$, then $T = g^{\mathrm{ord}(g)}$ is zero-sum and $\mathsf{v}_g(T) = \mathsf{v}_g(S)$. Thus every multiplicity is strictly smaller than the element order; in particular, $0 \nmid S$. If S is regular, Corollary 4.3 applies.

Otherwise choose H by Lemma 4.4. Then H is nontrivial, S_H is regular over H, and $|S_H| \ge |H|$. First suppose $H = C_p \oplus C_{mp}$ is noncyclic. If $m = 1$, the known value of $\mathsf{N}(C_p^2)$ is at most $p^2 \le |S_H|$. If $m \ge 2$ and $p \ge 5$, then $|S_H| \ge mp^2 \ge \mathrm{ub}(H)$, so Corollary 4.3, applied within H, suffices. For $p = 3$, use Lemma 4.8. In each case, $\mathsf{v}_h(S_H) = \mathsf{v}_h(S)$ for every $h \in H$, so the subsequence obtained in S_H also has the required property in S.

We may therefore suppose $H = C_m$ is cyclic. If $|S_H| \ge \mathsf{N}(H)$, apply its definition within H. Otherwise $m \le |S_H| \le \mathsf{N}(H) - 1$. Let $S_1 = SS_H^{-1}$, and let $\Phi : G \to G_1 = G/H$

be the quotient map. If G_1 is cyclic, apply Lemma 4.12 when $|G_1| = p$ and Lemma 4.11 otherwise. These lemmas apply because $|S| \geq \mathrm{ub}(G) \geq 2np - 1$.

Suppose finally that G_1 is noncyclic. Since $G_1/pG_1 \cong G/(H + pG)$ has order $p^2 = |G/pG|$, we obtain $H \leq pG$. Thus $1 < m \mid n$ and $G_1 \cong C_p \oplus C_{kp}$, where $k = n/m < n$. If $\mathsf{v}_{\bar{g}}(\Phi(S_1)) \geq \mathrm{ord}(\bar{g})$ for some $\bar{g} \in G_1 \setminus \{0\}$ and $|\Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)| \geq 2$, Lemma 4.5 applies. Otherwise, for each $g \mid S_1$ with $\mathsf{v}_g(S_1) \geq r_g = \mathrm{ord}(\Phi(g))$, choose $\lfloor \mathsf{v}_g(S_1)/r_g \rfloor$ disjoint copies of g^{r_g}. Let T_0 be their product, U the sequence of their sums, and $R = S_1 T_0^{-1}$ the remaining sequence. Every term of U is nonzero because $r_g \leq \mathsf{v}_g(S) < \mathrm{ord}(g)$, and $\Phi(R)$ is weak-regular. If $|U| \geq m - 1$, Lemma 4.10 applies with $K = H$. Otherwise $|T_0| \leq (m-1)kp$, and

$$|R| \geq 2np - 1 - (\mathsf{N}(H) - 1) - (m-1)kp \geq \mathsf{N}(G_1).$$

The last inequality is Lemma 2.6, using the appropriate induction bound for $k \geq 2$, or the known elementary-group value for $k = 1$. By the definition of $\mathsf{N}(G_1)$, there is $R_1 \mid R$ with $\sigma(\Phi(R_1)) = 0$ and $\mathsf{v}_{\bar{g}}(\Phi(R_1)) = \mathsf{v}_{\bar{g}}(\Phi(R))$ for some $\bar{g} \mid \Phi(R)$. Then $T_1 = R_1 T_0$ has sum in H and contains all terms of S_1 whose image is $\bar{g}$. In particular, $\mathsf{v}_h(T_1) = \mathsf{v}_h(S_1)$ for any $h \in \Phi^{-1}(\bar{g}) \cap \mathrm{supp}(S_1)$. Since $\sum_0(S_H) = H$, Lemma 2.8 completes the proof. Replacing n by np proves the asserted equality for $C_p \oplus C_{np^2}$. □

5. Remarks and open questions

We conjecture that $\mathsf{N}(C_p \oplus C_{np}) = 2np - 1$ whenever $n \geq 2$ and $p \geq 3$ is the smallest prime dividing np. The non-regular case in the proof of Theorem 1.4 uses only $|S| \geq 2np - 1$. Hence, for the remaining cases $p \nmid n$, the bound $c_0(C_p \oplus C_{np}) \leq \lceil (4np - 2)/3 \rceil$ would imply the conjecture by Proposition 4.2.

For the cyclic groups below, let $A_{n+1} = \lceil 2\sqrt{n+1} \rceil$ and

$$B_{n+1} = \min\{a + b : a, b \in \mathbb{N},\ ab \geq n + 1,\ \gcd(a, n) = 1\},$$

as in [8]. For the following two families, $\mathsf{N}(C_n)$ attains the upper bound $2n - A_{n+1} + 1$ and is strictly larger than $2n - B_{n+1} + 1$.

Theorem 5.1. *Let $n = 2730\big(1 + 105(2t-1)\,(13t - \frac{13\pm1}{2})\big)$ for a positive integer t; equivalently, $n \in T_2 \cup T_3$, where*

$$T_2 = \{\, 2730(2730t^2 - 2835t + 736) \,\}_{t \geq 1},$$
$$T_3 = \{\, 2730(2730t^2 - 2625t + 631) \,\}_{t \geq 1}.$$

Then

$$B_{n+1} = A_{n+1} + 1 = \lceil 2\sqrt{n+1} \rceil + 1,$$

and

$$\mathsf{N}(C_n) = 2n - B_{n+1} + 2 = 2n - \lceil 2\sqrt{n+1} \rceil + 1.$$

Proof. By [8, Lemmas 5.2 and 4.3], $B_{n+1} = A_{n+1} + 1$ and $\mathsf{N}(C_n) \leq 2n - A_{n+1} + 1$. Let $C_n = \langle e \rangle$.

Suppose first that $n \in T_2$, and let $d = 1365(2t-1) - 52$ and $x = (12n - d)/13$. Then $n = d(d-1) - 26$, $13 \mid d$, and d is odd. Since $d \geq 1313$, we have $(d-1)^2 < n+1 < (d - \frac{1}{2})^2$, and hence $A_{n+1} = 2d - 1$. The identity $n = d(d-1) - 26$ gives

$$13x \equiv -d \pmod{n}, \qquad (d-1)x \equiv -2 \pmod{n}.$$

Every common divisor of x and n therefore divides both d and 2, so $\gcd(x, n) = 1$. As $1 < x < n$, the sequence $S = e^{n-d}(xe)^{n-d+1}$ is weak-regular and has length $2n - A_{n+1}$. If a zero-sum subsequence contains all $n - d + 1$ copies of xe, its multiplicity of e is congruent to $(d-1)x \equiv -2 \pmod{n}$. It must therefore equal $n - 2 > n - d$, a contradiction. If a zero-sum subsequence contains all $n - d$ copies of e, then $13x \equiv -d \pmod{n}$ and $\gcd(x, n) = 1$ force its multiplicity of xe to equal $n - 13 > n - d + 1$, also a contradiction.

Suppose now that $n \in T_3$, and let $d = 1365(2t-1)+53$ and $x = (n-d)/2$. Then $n = d(d-1)-26$, d is even, and $d \equiv 1 \pmod{13}$. In particular, $d \geq 1418$ and $A_{n+1} = 2d-1$ as above. We have

$$2x \equiv -d \pmod{n}, \qquad (d-1)x \equiv -13 \pmod{n}.$$

Every common divisor of x and n divides both d and 13, so $\gcd(x,n) = 1$. Again $1 < x < n$, and $S = e^{n-d}(xe)^{n-d+1}$ is weak-regular with $|S| = 2n - A_{n+1}$. A zero-sum subsequence containing all copies of xe would have $n - 13 > n - d$ copies of e, by the second congruence. A zero-sum subsequence containing all copies of e would have $n - 2 > n - d + 1$ copies of xe, by the first congruence and $\gcd(x,n) = 1$. Thus S has no zero-sum subsequence T with $\mathsf{v}_g(T) = \mathsf{v}_g(S)$ for $g \in \{e, xe\}$. In both cases, the definition of $\mathsf{N}(C_n)$ gives $\mathsf{N}(C_n) \geq 2n - A_{n+1} + 1$, proving the equality. □

Remark 5.2. The construction of Theorem 5.1 relies on the specific relation $n = d(d-1)-26$ together with the arithmetic of 13 modulo d (or $d-1$). For the other two families T_4, T_5 identified in [8, Lemma 5.2] (which satisfy $n = d^2 - 64$ rather than $n = d(d-1)-26$), the analogous construction does not apply directly, and whether $\mathsf{N}(C_n)$ on $T_4 \cup T_5$ attains the upper or the lower bound remains an open question.

Acknowledgement. The author thanks W. Gao for his guidance and helpful suggestions. This work was supported in part by the National Natural Science Foundation of China (Grant No. 12071344). This paper is based on parts of Chapters 4 and 5 of the author's doctoral dissertation, completed in 2025.

References

1. W. Gao and A. Geroldinger, *Zero-sum problems in finite abelian groups: a survey*, Expo. Math. **24** (2006), 337–369.
2. W. Gao, D. Han, G. Qian, Y. Qu and H. Zhang, *On additive bases II*, Acta Arith. **168** (2015), 247–267.
3. W. Gao, S. Hong, W. Hui, X. Li, Q. Yin and P. Zhao, *Representation of zero-sum invariants by sets of zero-sum sequences over a finite abelian group*, Period. Math. Hung. **85** (2022), 52–71.
4. W. Gao, S. Hong, W. Hui, X. Li, Q. Yin and P. Zhao, *Representation of zero-sum invariants by sets of zero-sum sequences over a finite abelian group II*, J. Number Theory **241** (2022), 738–760.
5. W. Gao, Y. Li, Y. Qu and Q. Wang, *Additive bases of finite abelian groups of rank 2*, Integers **22** (2022), Paper No. A37.
6. W. Gao, Y. Qu and H. Zhang, *On additive bases III*, Acta Arith. **193** (2020), 293–308.
7. A. Geroldinger and F. Halter-Koch, *Non-Unique Factorizations. Algebraic, Combinatorial and Analytic Theory*, Pure and Applied Mathematics, Vol. 278, Chapman & Hall/CRC, Boca Raton, FL, 2006.
8. X. Jiang and W. Yang, *A zero-sum problem related to the max gap of the unit group of the residue class ring*, J. Number Theory **260** (2024), 198–211.
9. M. B. Nathanson, *Additive Number Theory: Inverse Problems and the Geometry of Sumsets*, Graduate Texts in Mathematics, Vol. 165, Springer, New York, 1996.
10. Y. Qu and D. Han, *Additive bases of $C_p \oplus C_{p^n}$*, Int. J. Number Theory **13** (2017), 2453–2459.
11. Y. Qu and Y. Li, *Additive bases of $C_3 \oplus C_{3q}$*, Colloq. Math. **169** (2022), 189–195.
12. Y. Qu, Y. Li, Q. Wang and X. Zhao, *A note on additive bases of abelian groups of rank 2*, Graphs Combin. **41** (2025), Paper No. 24.